Sentiment Analysis of Arabic Song Lyrics:

A Case Study on George Wassouf's Discography

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Abstract

Understanding the emotional tone of Arabic song lyrics offers insight into cultural and artistic expression. This work presents a computational pipeline for analysing the lyrics of George Wassouf's discography, a dataset of 78 songs by 17 composers and 34 lyricists. The pipeline normalises Arabic text, scores sentiment with a pretrained dialectal-Arabic transformer (CAMeLBERT), extracts recurring terms with TF-IDF, and represents each song with AraBERT embeddings that are clustered with K-Means and visualised with UMAP. The model labels 65.4% of the songs as negative, 20.5% as positive and 14.1% as neutral, consistent with the themes of separation and longing in the repertoire. Because most composers and lyricists appear in only one or a few songs, per-collaborator results are exploratory. The full implementation is available on GitHub.

1. Introduction

Arabic music, exemplified by the work of George Wassouf, is known for emotionally rich lyrics about love, longing and heartbreak. Measuring the sentiment of such lyrics can reveal patterns in artistic expression. Classical sentiment methods struggle with Arabic because of its rich morphology, right-to-left script, optional diacritical marks and wide dialectal variation, and song lyrics are mostly written in dialect rather than Modern Standard Arabic. Transformer-based models pretrained on Arabic, including dialectal data, make contextual analysis of such text practical.

This paper proposes a pipeline that (i) cleans and normalises the lyrics, (ii) predicts sentiment for every song, (iii) extracts the most characteristic terms, and (iv) maps songs into a semantic space to find similar songs and groups. We also compute sentiment aggregates by composer and lyricist to explore collaboration patterns. The pipeline is designed to be reusable for other Arabic artists.

2. Proposed Method

2.1 Dataset

The dataset contains 78 songs. Each record has the track title, lyrics, release year, composer and lyricist. The songs involve 17 distinct composers and 34 distinct lyricists.

2.2 Preprocessing

Preprocessing makes the text compatible with NLP tools and consists of: (a) removing punctuation, line breaks and Arabic diacritics (Tashdid, Fatha, Kasra and others) with regular expressions; (b) normalising letter variants (alef forms to bare alef, final ya to alef maqsura, final ta marbuta to ha); (c) extracting Arabic-only track titles by removing Latin characters and hyphens; (d) removing stopwords with NLTK's Arabic list; and (e) keeping all text in UTF-8. After cleaning, a song contains on average about 340 words (median 294).

2.3 Sentiment Analysis

Sentiment is predicted with the dialectal-Arabic variant of CAMeLBERT (*CAMeL-Lab/bert-base-arabic-camelbert-da-sentiment*) from the Hugging Face library [3]. The model returns one of three labels (positive, neutral, negative) with a confidence score. Inputs are tokenised with the model's tokenizer and truncated to the 512-token limit, and the predictions are stored alongside each song's metadata.

2.4 Keyword Extraction

TF-IDF is computed over the preprocessed lyrics to find the terms that contribute most to the catalogue.

2.5 Embeddings, Clustering and Visualisation

Each song is encoded with AraBERT v2 [2] to obtain a 768-dimensional embedding. Cosine similarity between embeddings gives, for each song, its five most similar songs. Songs are grouped with K-Means (k = 4), and the embeddings are projected to two dimensions with UMAP [4] for visualisation. Arabic labels are shaped and reordered with *arabic-reshaper* and *python-bidi* so they render correctly right-to-left.

2.6 Composer and Lyricist Analysis

Sentiment scores are aggregated per composer, per lyricist and per lyricist-composer pair to explore whether collaborations differ in tone.

3. Results

3.1 Sentiment Distribution

Sentiment predictions were obtained for all 78 songs (Table 1, Figure 1). The catalogue is predominantly negative, reflecting themes of heartbreak, separation and nostalgia. For example, the song *Al-Forqa Sa'ba* (The Separation Is Hard) is classified as negative with a confidence of 0.94.

Sentiment	Songs	Share
Negative	51	65.4%
Positive	16	20.5%
Neutral	11	14.1%
Total	**78**	**100%**

Table 1. Predicted sentiment of the 78 songs.

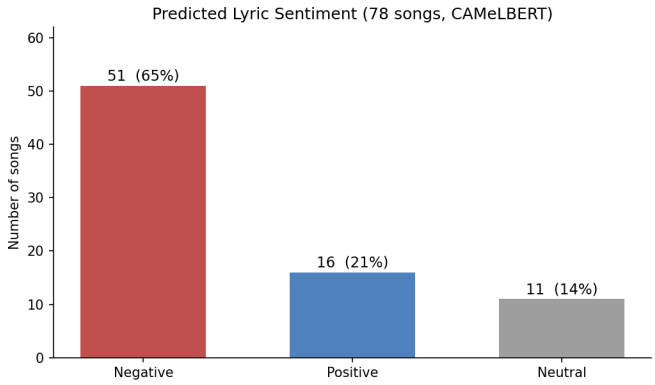


Figure 1. Distribution of predicted sentiment across the catalogue.

3.2 Recurring Terms

The highest-scoring TF-IDF terms are intimate and emotional: *qalbi* (my heart), *habibi* (my beloved), *al-hawa* (passion), *al-dunya* (the world), *layl* (night), *rohi* (my soul) and *eini* (my eyes) (Figure 2). Dialectal function words such as *fi*, *illi* and *ana* also rank highly because the NLTK stopword list targets Modern Standard Arabic; a dialect-aware stopword list would remove them.

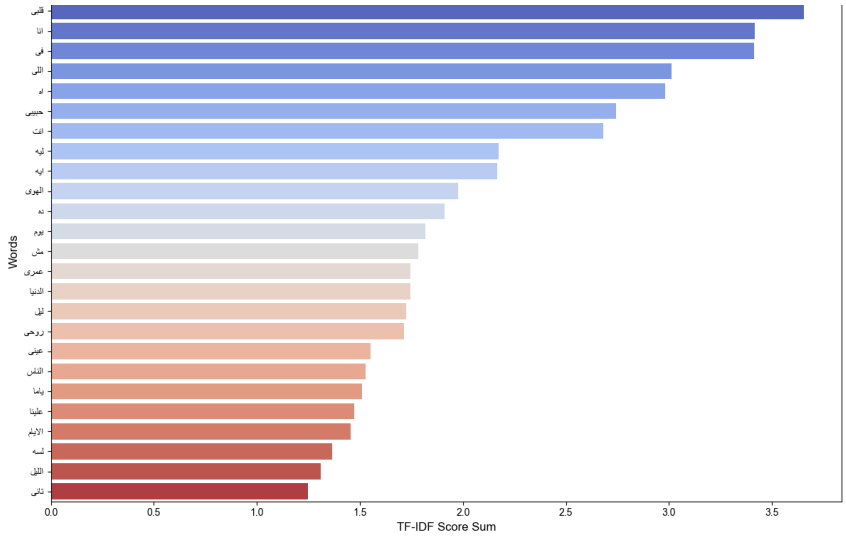


Figure 2. Top TF-IDF terms in the lyrics.

3.3 Song Space

Figure 3 shows the UMAP projection of the song embeddings coloured by K-Means cluster. Songs that are close together share vocabulary and style, and the similarity table lists the five nearest songs for each track. The clusters are not given semantic names; interpreting them is left for future work.

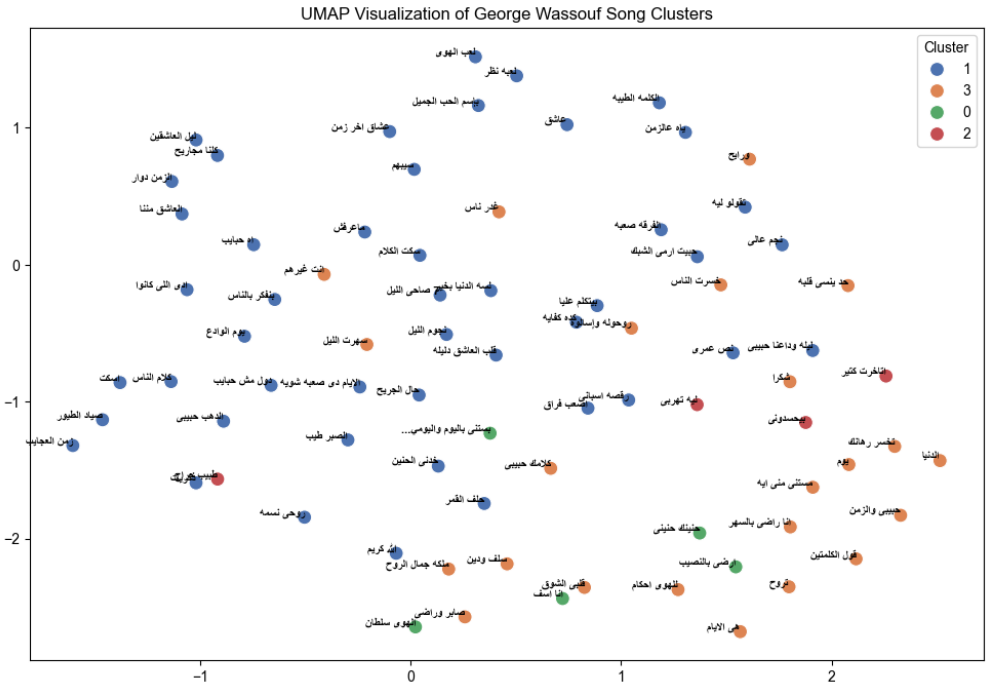


Figure 3. UMAP projection of song embeddings coloured by K-Means cluster (k = 4).

3.4 Composers and Lyricists

Average sentiment was computed for each composer, lyricist and collaboration pair. However, most collaborators appear in only one to three songs, so a pair's average often reflects a single song, and correlations between composers computed on such sparse data are not meaningful. We therefore report this analysis as exploratory and draw no conclusions about individual collaborators.

4. Limitations

The study covers a single artist and 78 songs, so the findings describe this catalogue rather than Arabic music in general. Songs are written in dialect while many tools are designed for Modern Standard Arabic. Transformer models accept at most 512 tokens and many lyrics are longer, so long songs are truncated; splitting lyrics into chunks and aggregating the predictions would be more faithful. The model labels have not been validated against human annotations.

5. Conclusion and Future Work

We presented a pipeline for analysing Arabic song lyrics and applied it to George Wassouf's discography. Handling Arabic-specific challenges (diacritics, letter variants, right-to-left rendering) allowed reliable sentiment labelling, keyword extraction and a semantic map of the catalogue. Future work includes chunking long lyrics, validating labels with human annotators, adding more artists to make collaboration analysis statistically meaningful, and more nuanced emotion models for dialectal Arabic.

References

[1] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of NAACL-HLT*, 4171-4186.

[2] Antoun, W., Baly, F., and Hajj, H. (2020). AraBERT: Transformer-based Model for Arabic Language Understanding. *arXiv preprint arXiv:2003.00104*.

[3] Inoue, G., Alhafni, B., Baimukan, N., Bouamor, H., and Habash, N. (2021). The Interplay of Variant, Size, and Task Type in Arabic Pre-trained Language Models. *Proceedings of the Sixth Arabic Natural Language Processing Workshop (WANLP)*.

[4] McInnes, L., Healy, J., and Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. *arXiv preprint arXiv:1802.03426*.